



THE UNIVERSITY OF HARIPUR

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WEEK 6 INTERNSHIP PROGRESS REPORT

Remote Internship at FlyRank AI

STUDENT INFORMATION

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Department	Department of Information Technology
Program	BS Artificial Intelligence
Semester	7th Semester
Company	FlyRank AI
Internship Type	Remote Internship
Internship Duration	July 2026 – August 2026 (8 Weeks)
Report Period	Week 6 (August 5 – August 11, 2026)

Submitted To:

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1. Introduction

FlyRank AI is an AI-powered organic growth platform that helps businesses improve their visibility across traditional search engines and AI answer engines such as ChatGPT, Perplexity, and Gemini. The platform automates large parts of the content lifecycle - research, scoring, and optimization - primarily serving D2C, SaaS, and agency clients who need scalable organic growth.

This report covers Week 6 of my remote Machine Learning Engineering internship at FlyRank AI, under the Applied Search Intelligence track. It summarizes the assignment completed, the reasoning behind it, and the skills developed.

2. Objective of Week 6

Objective: To audit my own Week 5 model the same way a research paper's methodology would be audited - applying the 'attack your own model' rigor from this week's live session, which used FlyRank's own published research paper as the worked example.

3. Week 6: Validation Audit & Model Honesty Check

Task Requirements: Select two findings from FlyRank's research paper and write a constructive methodology question for each; re-run my Week 5 model under an honest split (grouped by client) with a clear before/after comparison; audit my own features for leakage; review real failure examples; rewrite my own earlier claims into public-safe language.

Research Performed: Read FlyRank's 'State of AI-Driven SEO' research paper in full, paying particular attention to its Health Score formula, its ML appendix feature-importance methodology, and its own stated limitations section, to identify two findings worth a constructive methodology question.

Summary of Work Completed: Named two paper findings and posed a respectful methodology question for each: (1) whether word count and age were measured before or during the observed growth window in the 'Anatomy of Growing Content' finding, and (2) whether the Health Score feature-importance ranking remains meaningful given that Average Position and Impressions are direct arithmetic components of the Health Score label itself. Re-ran my own Week 5 Random Forest model under both a random split and a grouped (client-holdout) split, reporting both Precision@50 numbers side by side - the honest, grouped-split score came out to 0.540, clearly lower than the random-split number, and that gap was itself treated as a finding about memorization. Ran a full leakage-audit checklist, including a deliberate leak test: adding a label-derived column (trend_pct) pushed the score to a near-perfect 1.000, confirming the test harness correctly detects leakage when present. Rewrote three of my own earlier claims from Week 4 and Week 5 into public-safe, decision-support language following the claim-ladder principle.

What Was Learned: Learned that a model's score means nothing without a same-split, same-metric baseline sitting next to it, and that the gap between a dishonest and an honest split is itself a measurable finding. Also learned to review published work constructively - asking 'how would this claim get stronger' rather than looking for mistakes - and to apply that same standard to my own earlier notebooks.

4. Skills Developed

Week 6 contributed to the development of the following technical, analytical, and professional skills:

Skill	Description
Methodology Review	Critiqued a real research paper's evidence chain constructively, identifying a label-derived feature issue in its own ML appendix.
Honest Validation Design	Re-ran a model under both a naive and a grouped split, quantifying the memorization gap between them.
Leakage Auditing	Executed a full attack-your-own-model checklist, including a deliberate leak test to verify the test harness itself works.
Claim Calibration	Rewrote prior claims to match the claim ladder - observed, directional, decision-support language only.

5. Challenges Faced

- › Framing a critique of a real, published paper respectfully and constructively rather than as a 'gotcha', while still being technically precise about the label-derived feature concern.
- › Designing the deliberate leak test so it clearly proved the test harness would catch leakage if present, not just that removing a feature changed the score.

6. Knowledge and Experience Gained

Week 6 reinforced that rigor is a practice applied equally to others' work and one's own - and that a validated, honestly-split result is the only kind of result worth building further recommendations on.

7. Conclusion

Week 6 turned the audit lens both outward, onto a real published research paper, and inward, onto my own Week 5 model. The honest grouped-split evaluation and the leakage-audit checklist together confirm the model's result is real signal, not memorization or leakage.

Looking ahead, the objective for the next reporting period is to turn this validated, honest model output into a practical, human-reviewed content action playbook.